

Introduction

Why are people interested in investing into stocks? What drives investors to invest in the first place? Valuation is the determining factor for investing into a stock. Valuation determines how much the market thinks a company is worth relative to what it actually produces. Whether investors are looking at EV, EBITDA, and comparing them to one another, what matters in the end is whether investors see the potential for the *future* not for the *present*. Let's speak about two companies ~ NVIDIA (NVDA) and compare it to Alphabet (GOOG). Nvidia and Alphabet are both pioneers in their respective fields, leading to continuous stock growth mainly because of the desire to be a part of both companies future success. Nvidia has excelled the past few years, consistently growing 50% plus percent YoY, as the time for AI development is now. Alphabet on the other hand, has been excelling in their practical use of AI. As a customer of Nvidia, Alphabet's current AI model (Gemini) has seen much success. However, Alphabet is still finding its way in the AI race as they integrate their products with AI. What is most important to pay attention to is the conversation of whether Nvidia deserves a valuation of \$4.4 trillion, as it depends on a very important yet deciding question - "Will they grow more in the future?". This is the deciding question for all stocks, but is crucial for Nvidia because of their involvement in today's global race to AI empowerment. Nvidia currently holds the highest market cap of any publicly traded company in the world, but is the \$4 trillion something worth worrying about? Nvidia's premium valuation reflects their extraordinary growth inside the financial statements; mainly driven by AI infrastructure demand from Alphabet, Microsoft, Meta, and more, instead of irrational investing.

Financial Statements Overview

Through the financial statements of both companies, we are able to tell which company is being bet on to win in the end. Through P/E (Price-to-Earnings), EV/EBITDA (Enterprise Value to Earnings Before Interest, Taxes, and Depreciation & Amortization) and P/S(Price to Sales), we are able to come up with a conclusion. Firstly, Price-to-Earnings tells us for every dollar a company earns in profit, how much investors are willing to pay for that profit. Nvidia's P/E sat at around 37x, meaning that investors are paying \$37 for every \$1 of earnings. Alphabet holds a P/E of 28x, meaning that for every \$1 the company makes, an investor is paying \$28 for that \$1 in earnings. Generally, the higher the P/E, the more confidence investors have in future earnings growth. Secondly, EV/EBITDA tells us the P/E valuation for the whole company, including debt. It is useful because it strips out differences in how companies are financed or taxed, making comparisons cleaner. Nvidia has their EV/EBITDA at around 33x, whereas Alphabet has their EV/EBITDA at around 24x. Nvidia carries the premium in this as well, as investors' ambitions for the company rise on a daily basis. The last notable measurable valuation comparison lies in P/S (Price-to-Sales), where for every dollar of revenue a company generates, what is the marketing valuing it at? This is most useful when comparing companies at different stages of profitability. Nvidia is currently at 20x, whereas Alphabet is at 9x. This precisely shows how much the market values each dollar of Nvidia's revenue. As a result of these 3 multiples, they are clearly telling us that investors are paying a premium for Nvidia's *future*, and not so much its present value.

Reason for Nvidia's High Premium

Why are people paying so much for the future value, as opposed to Alphabets? The answer doesn't directly undermine Alphabet's ability to succeed in the future, but more importantly where the companies stand in the current market, and how investors believe that market will evolve. As of now, Nvidia is the primary distributor for GPUs. GPUs are proficient in training AI, the very thing every

company is doing as of now. In response to companies needing GPUs to evolve their models as best as they can, they are quickly buying from GPU giants like Nvidia. This is evident in the growth of Nvidia over the past few years. From FY2024, Nvidia was reporting around \$60,922 (million), from FY2025, Nvidia was reporting around \$130,497 (million), and from FY2026, Nvidia reported a revenue of \$215,938(million). The reason investors are so interested is because from FY2024 to FY2025, Nvidia's revenue increased about 114%, and from FY2025 to FY2026 Nvidia's revenue increased 65%. Even though Alphabet has higher revenues and is a more established company, what drives investors to invest in a company is due to the fact that the company is significantly growing. Alphabet only grew about 15% YoY, and clearly investors aren't paying as much of a premium for Alphabet's growth compared to Nvidia. Investing as a whole, is mostly always a hypothesis on the future, where you use research and current trends to predict whether or not a company will be useful in the future, so that you can grab a piece of their future revenue ~ the reason why people invest. Even though the high multiples for Nvidia are what people are willing to pay for the stock, it still doesn't necessarily mean the stock is "overvalued" or that people are buying "too much of it". It is mainly due to the fact that the company is currently in the middle of one of the biggest technological advancements in recent history, and is the number one distributor in it.

Nvidia's Revenue Explained

Nvidia's business plan/revenue isn't imaginary, or lucky, it is in fact reliable and adaptable. Their involvement in data centers and AI chips make them extremely competitive, and often the winner of this volatile market. Nvidia GPUs are the core piece of hardware for every major AI company, cloud provider, and government that needs to train and run AI models. Microsoft, Alphabet, Amazon, Meta are all buying Nvidia chips on a massive scale. That is why they are reliable, because they are in constant demand. What makes their business plan adaptable is that Nvidia isn't just selling hardware, but they are empowering their software ecosystem through their chips called CUDA, which is a software platform developers use to program Nvidia GPUs. AI developers have built everything on CUDA over the last decade, so switching to another competitor's chips isn't just expensive, but requires rewriting enormous amounts of code. This is a real game changer that makes their revenue durable, and not just rely on their hardware to do the heavy lifting. The interesting intersection between Alphabet and Nvidia, is that Alphabet is both a customer and a competitor. Alphabet buys Nvidia chips for its cloud infrastructure, but they're also developing their own AI chips called TPUs (Tensor Processing Units), to reduce the dependence on Nvidia. This shows that the demand is real, and that the big competitors are trying to replicate what Nvidia is doing, which pushes Nvidia to do better overall - for now.

Nvidia's Risks

But this thesis has conditions attached to it, as there are still major risks for Nvidia. These risks mainly include the scenario in which companies aren't interested in their chips anymore, affecting their P/E, competition from other companies like AMD, and Geopolitical risks. The question of whether Nvidia will be dependable in the future is complex, as the company is cyclical by nature. If the AI buildout slows, where major tech companies begin to reduce the amount of chips from Nvidia they buy, Nvidia's revenue will drop sharply. However, Nvidia isn't a stranger to this; in 2022, Nvidia's gaming chip business collapsed almost overnight when pandemic-era demand dried up and inventory piled up, where their stock dropped over 60%. The AI chip dependence is much bigger, but the same logic applies. If at some point major cloud providers choose to slow down on spending towards GPUs because they are no

longer needed, Nvidia's revenue growth can certainly begin to go negative. When a stock is priced 37x their earnings based on growth expectations, a sudden slowdown doesn't just hurt the business, but destroys the P/E multiple as well. Earnings will fall, and investors are no longer willing to pay a premium for ownership, because they aren't expecting any significant future growth. The other risk lies in their competitors. Although Nvidia has done well selling their chips and leading the GPU industry, competitors like AMD are not backing down. AMD's business plan relies heavily on making chips overall cheaper, and also making custom chips for companies like Amazon and Alphabet. AMD can gradually reduce dependence on Nvidia chips, but it is no easy task, and will certainly not happen overnight. The last risk worth noting is their geopolitical risk, especially with China. The US government has restricted Nvidia from selling its most advanced chips to China. China was a meaningful revenue source and those sales are now blocked or severely limited. To combat this, Nvidia designed a watered-down chip (H20) specifically for China to work around those restrictions, and the government still restricted those chips. This is an ongoing risk that has nothing to do with demand.

Conclusion

So why is Nvidia valued so highly? It is tough to say whether or not the company should be valued as highly as it is, but Nvidia does speak for itself, and they back it up. Their continuous growth ~ 65% YoY, and their adaptable business model does show signs of a meaningful and reliable business. However, the risks they face are serious, and if left unchecked, can lead to the company and its shares to sharply decline. So, does Nvidia deserve its \$4.4 trillion valuation? Based on the numbers and current positioning of the company during this AI wave, Nvidia's premium appears justified for as long as AI infrastructure demand remains at its current pace. However, whether or not the company is worth investing in depends on your perspective of the AI industry, and whether or not you believe that companies will keep relying on this technology and if they will need Nvidia to advance their ideas.